OXFORD JLB JOURNAL OF LEUKOCYTE BIOLOGY

Journal of Leukocyte Biology, 2024, 116, 1385–1397
https://doi.org/10.1093/jleuko/qiae180
Advance access publication 19 September 2024
Article

Rapid progression of CD8 and CD4 T cells to cellular exhaustion and senescence during SARS-CoV2 infection

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Abstract
Risk factors for the development of severe COVID-19 include several comorbidities, but age was the most striking one since elderly people were disproportionately affected by SARS-CoV-2 infection. Among the reasons for this markedly unfavorable response in the elderly, immunosenescence and inflammaging appear as major drivers of this outcome. A finding that was also notable was that hospitalized patients with severe COVID-19 have an accumulation of senescent T cells, suggesting that immunosenescence may be aggravated by SARS-CoV-2 infection. The present work was designed to examine whether these immunosenescence changes are characteristic of COVID-19 and whether it is dependent on disease severity using cross-sectional and longitudinal studies. Our cross-sectional data show that COVID-19, but not other respiratory infections, rapidly increased cellular senescence and exhaustion in CD4 and CD8 T cells during early infection. In addition, longitudinal analyses with patients from Brazil and Portugal provided evidence of increased frequencies of senescent and exhausted T cells over a 7-d period in patients with mild/moderate and severe COVID-19. Altogether, the study suggests that accelerated immunosenescence in CD4 and especially CD8 T-cell compartments may represent a common and unique outcome of SARS-CoV2 infection.

Keywords: COVID-19, immunosenescence, inflammaging, inflammatory cytokines, T-cell exhaustion, T-cell senescence

1. Introduction

Since 2019 when it first started in China, COVID-19 pandemic caused by the severe acute respiratory syndrome coronavirus 2 (SARS-CoV2) has been responsible for the death of more than 6 million people worldwide.[1] Clinical manifestations of COVID-19 range from asymptomatic to severe pneumonia associated with extensive immune–inflammatory response, a clinical condition that can lead to multiorgan failure and prolonged complications.[2] Although most of infected individuals are asymptomatic or develop mild symptoms, disease severity and mortality rate can change between regions, depending on the characteristics of the population as well as the adoption or not of mitigation strategies to slow the spreading of the disease. The European population presented a high mortality rate when compared with other developed countries regardless of a lower prevalence of comorbidities, possibly due to a greater number of elders in the population.[3,4] On the other hand, the elevated mortality rates observed in Latin America have been associated with the lack of public health policies aiming the containment of the disease during the pandemic surge.[5]

Risk factors for the development of severe COVID-19 include obesity, diabetes, high blood pressure, male sex, and age, with the latter being the most striking, since elderly people were disproportionately affected by SARS-CoV-2 infection.[6–10] Among the reasons for this markedly unfavorable response in the elderly, immunosenescence and inflammaging appear as major drivers of this outcome.[11,12] Immunosenescence is the aging of the immune system and is mainly characterized by a decrease in naïve T-cell numbers together with an accumulation of $CD4^+$ and $CD8^+$ memory and terminal effector T cells, resulting in increased vulnerability to infections and impaired response to vaccination.[13–15] Moreover, aged T lymphocytes, like other senescent cells, tend to develop a senescence-associated secretory phenotype (SASP), characterized by the production and secretion of IL-6, IL-1 β, IL-8, IL-18, and TNF along with other inflammatory mediators. Senescent cell accumulation and other aging-related disturbances (mitochondrial dysfunction, dysbiosis of microbiota, innate immune responses) lead to a chronic low-grade inflammatory state known as inflammaging.[14,16,17]

Continuous immune dysregulation induced by prolonged exposure to infectious and noninfectious antigens can also drive immune cell aging, since the accumulation of inflammatory mediators secreted in response to consecutive infections further

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Received: November 24, 2023. **Revised:** May 19, 2024. **Corrected and Typeset:** September 19, 2024

contributes for the impairment of the adaptive immune response, boosting immunosenescence and inflammaging.[14,18,19] Chronic infections, including those caused by the human cytomegalovirus (CMV), human immunodeficiency virus (HIV), hepatitis B virus (HBV), and human papillomavirus (HPV), are sources of constant stimulation to the immune system, resulting in the accumulation of terminally differentiated senescent T-cell populations and, consequently, immunosenescence.[20,21] Similarly, although it is not a persistent infection, SARS-CoV2 can also trigger the accumulation of exhausted and senescent T-cell phenotypes due to its extensive activation and clonal expansion.[22,23] Recent publications have shown that patients with severe COVID-19 presented an increased frequency of terminally differentiated CD4 and CD8 T cells when compared with healthy controls matched for sex and age.[24,25] However, it is not clear whether this is unique to COVID-19 or acute infections caused by other viruses, such as influenza or other coronaviruses, have the same impact on T-cell differentiation. Moreover, there is no longitudinal study clearly demonstrating that SARS-CoV2 infection leads to accelerated immunosenescence as a general phenomenon in distinct populations and in different forms of COVID-19.

In this study, we investigated the evolution of the inflammatory response and immunosenescence in patients with COVID-19 and flu-like syndrome (FLS), followed by a longitudinal analysis of patients diagnosed with COVID-19 from Belo Horizonte, Brazil, and Lisbon, Portugal. Our results showed that SARS-CoV2 infection was associated with progression to T-cell senescence and exhaustion when compared with FLS caused by other viruses in patients examined at two time points of infection. In longitudinal analyses, this outcome was confirmed in both mild and severe cases of COVID-19 in cohorts of individuals from Brazil and Portugal.

2. Materials and methods

2.1 Ethics statement

The research protocol was approved by the National Research Ethics Commission (CONEP # 5.190.260) of Brazil and by the Lisbon Academic Medical Center Ethics Committee (ref. no. 306/20). The study was conducted in accordance with the Helsinki Declaration for research involving humans. All participants agreed to participate voluntarily in this study without financial support and signed an informed consent.

2.2 Study design

A cross-sectional study was conducted with a total of 149 individuals from Belo Horizonte/MG, Brazil, recruited from December 2020 to October 2021. The SARS-CoV2 variants circulating during the patient collection period were the original one and P1. Volunteers were recruited in Belo Horizonte, Minas Gerais, at the Emergency Care Unit of the Centre-South region (UPA-CS), and at Hospital Risoleta Tolentino Neves. All volunteers presented mild symptoms, and they underwent a real-time polymerase chain reaction (RT-PCR) test to confirm SARS-CoV2 infection. Those presenting similar symptoms with a negative RT-PCR for SARS-CoV2 infection were classified as having a FLS with an unclear etiology.

The longitudinal study was performed at two healthcare centers: Centro Hospitalar Universitário Lisboa Norte in Lisbon, Portugal, and Hospital Risoleta Tolentino Neves in Belo Horizonte, Brazil. Nonvaccinated patients were recruited within the first 3 d of hospitalization or while in ambulatory care, after presenting a positive nasopharyngeal swab test for SARS-CoV2 by RT-PCR performed upon admission at the healthcare units. A total of 10 infected patients from Lisbon and 8 from Belo Horizonte agreed to participate in this study. For the longitudinal follow-up, blood samples were collected at the time of enrollment (T0) and 7 d later (T7). Each city had a control group formed according to the following criteria: testing negative for SARS-CoV2 in the RT-PCR test and not having presented any symptoms the week before blood sample collection. Individuals with prior inflammatory diseases, such as cancer and renal failure, were excluded from the study. Data collection was carried out between August and October 2020 in Lisbon and June and August 2021 in Belo Horizonte.

Disease severity classification was based on the WHO criteria.[1] Asymptomatic patients or those treated in an outpatient clinic were considered as mild cases. Moderate cases included hospitalized patients without oxygen support or using low- to moderate-flow oxygen by mask or nasal prongs. Severe cases were characterized as the need for high-flow oxygen, mechanical ventilation, vasopressors, dialysis, or extracorporeal membrane oxygenation.[1]

2.3 Blood collection and isolation of PBMCs

Peripheral blood mononuclear cells (PBMCs) were obtained by blood collection from patients with COVID-19 and healthy controls using heparinized vacuette tubes. The purification was attained by Ficoll gradient (Histopaque-1077; Sigma, cat#10771) centrifugation at room temperature, in a 1:2 ratio, for 40 min, at 600 × g and without brake-induced end-centrifugation deceleration phase. Then, PBMCs were collected and washed with Roswell Park Memorial Institute 1640 basal medium. Red blood cells were lysed using a lysis buffer. Cells were counted and stored in fetal bovine serum supplemented with 10% dimethyl sulfoxide at −80 °C until further use. Blood serum was also obtained by centrifuging whole blood collected in sera tubes for 10 min at 500 × g.

2.4 Immunophenotyping by polychromatic flow cytometry

PBMCs (1 × 106) were first labeled the cells with LIVE/DEAD fixable aqua dead cell marker (ThermoFisher Scientific, cat# L34957) and antibodies fluorophore-conjugated human to surface markers. anti-CD8 (RPA-T8, cat# 562282), anti-CD4 (SK3, cat# 557852), anti-CD25 (M-A251, cat# 555432), anti-CD278 (ICOS; DX29, cat# 562833), anti-CD57 (NK-1, cat# 563896), anti-CD279 (PD-1; EH12.1, cat# 563245), anti-TIGIT (741182, cat# 747840), anti-CD28 (CD28.2, cat# 561368), anti-CD45RO (cat# 304234), anti-CD3 (cat# 317340), anti-CCR7 (cat# 353216), anti-KLRG1 (cat# 138427), anti-CD16 (cat# 555408), anti-CD56 (cat# 340723), CD69 (cat# 557756), and anti-NKG2D (cat# 558071).

After the surface staining, cells were fixed and permeabilized using the Foxp3/Transcription Factor Staining Buffer Set from eBioscience (cat# 00-5523-00) and stained for -Foxp3 from ImmunoTools (3G3, cat# 21276106). Single-color controls for fluorescence compensation were prepared with antibody capture compensation beads (BD Biosciences). Cell samples were acquired in a BD LSR Fortessa cell analyzer (BD Biosciences) coupled to computers with DIVA and FlowJo-10 software (Tree Star).

2.5 Luminex-Multiplex measurement of serum cytokines, chemokines, and growth factors

A Bio-Rad Laboratories kit (Bio-Plex Pro Human Cytokine Standard) was used to analyze multiple serum mediators simultaneously with the Bio-Plex 200 system from Bio-Rad, following storage and processing protocols standardized by the Laboratory of Biomarkers (IRR-FIOCRUZ/MG). Samples were transported and stored at a temperature of −80 °C.

Analyses were performed using Bio-Plex xPONENT version 3.1 software (Bio-Rad) and included the following panel of analytes: IL-1β, IL-1ra, IL-2, IL-4, IL-5, IL-6, CXCL-8 (IL-8), IL-10, IL-12p70, IL-13, GM-CSF, IFN-γ, CCL-2 (MCP-1), and TNF. The panel of biomarkers included in the kit and their standard settings is presented in Supplementary Table 1.

2.6 Statistical analysis

Statistical analyses were performed using GraphPad Prism 8.0 software (GraphPad Software, San Diego, CA, USA). All data acquired from Luminex-Multiplex were converted to logarithmic scale in order to normalize the data, and a lognormality test was performed to confirm data normality. Wilcoxon's test was applied in nonparametric data to compare samples of the same patient at T0 and T7. Student's t-test was used for analyzing parametric data.

Radar plots highlight the contribution of different cytokines to immunological profile. This analysis allows conversion of quantitative cytokine measurements into a categorical variable of low and high cytokine producers as previously proposed by Luiza-Silva et al.[26] Briefly, this approach categorizes each subject as "Low" or "High" cytokine producer, taking the global median value as a specific cutoff edge for each cytokine. For the calculation of the global median, the whole universe of data obtained for the groups was considered. Following data categorization, the frequency of "high cytokine producers" was calculated for each group. On radar charts, each axis represents the percentage (%) of volunteers categorized as "high cytokine producers" in each group. Then, connecting the values of each axis forms a central polygonal area that represents the immunological profile. Relevant productions were considered when the percentage of "high cytokine producers" of a given cytokine was >50%. The values from each axis can be joined to form a central polygonal area that represents the overall proinflammatory or regulatory balance. The increase or decrease in the central polygonal area reflects a greater or lesser contribution of the proinflammatory or regulatory profile in each group.

Statistical analysis of data on frequencies of T-cell subsets obtained by flow cytometry was performed using the nonparametric Wilcoxon rank-sum test. $P < 0.05$.

3. Results

3.1 Characteristics of the study population

In the cross-sectional study, we evaluated included 149 individuals with either FLS or different forms of COVID-19, assessed at different times of the infection (Table 1). Participants were divided according to time of infection calculated by the reported appearance of the first symptoms: individuals who had flu-like symptoms for less than 4 d and those who had flu-like symptoms for 5 to 7 d. They were subsequently swab tested for SARS-CoV2 by RT-PCR. The mean age among participants with FLS was 51.0 (21 to 79) and 40.5 (21 to 73) yr for FLS 1 to 4 and FLS 5 to 7, respectively. Participants positive for COVID-19 with four or fewer days of symptoms had an average age of 47.5 (20 to 89) yr; for participants with 5 to 7 d of symptoms, the average was 45 (27 to 67) yr (COVID 5 to 7). Hypertension was the most frequent comorbidity among the groups, and mild COVID-19 presented a higher frequency of 60% of the evaluated individuals.

The longitudinal study included eight individuals from Belo Horizonte with mild to moderate cases of COVID-19 and 10 individuals from Lisbon with severe cases of COVID-19, assessed at admission (day 0) and 7 d after recruitment (day 7) (Table 2). The mean age of individuals in the infected groups was 60 (45 to 85) and 38 (35.7 to 56.8) yr in Lisbon and Belo Horizonte, respectively. The prevalence of comorbidities was higher in the Portuguese population, and hypertension was the most prevalent among infected patients. Deaths and missing samples were only registered in Lisbon.

3.2 Identifying the inflammatory profile of patients

To characterize the inflammatory profile of the study populations, we measured plasma levels of 27 cytokines, chemokines, and immune-related molecules. In the 149 patients of the cross-sectional study, the global production of inflammatory mediators of control patients with FLS and patients with COVID-19 was compared. Radar plots (Fig. 1A) show that both groups had an overall increase in the frequency of serum mediators after 5 d of infection, confirming the ongoing response to infection. When comparing the groups of FLS 1 to 4 and FLS 5 to 7 (Fig. 1B to M), a rise in serum levels of TNF, IL-15, 1L-17A, IL 4, IL-5, IL-9, VEGF, G-CSF, GM-CSF, CCL3, CCL4, and IL-1β. This suggests an intensification of the immune response throughout the analyzed infection period. Interestingly, during COVID-19 infection, as already described in the literature,[27] an outburst of mediators called cytokine storm occurred in the first days of infection. High levels of immune mediators were detected in the serum already in the first days of infection, and they persisted until the seventh day. In the first 4 d of SARS-CoV2 infection, there was increased production of CCL2, CLL4, IFN-γ, IL-4, VEGF, and IL-17 and of IL-10, IL-12p70, and GM-CSF after 4 d of infection (Fig. 1).

Table 1. Clinical features of individuals from Brazil with either FLS or COVID-19.

	Flu-like syndrome (1 to 4 d) ($n = 31$)	Flu-like syndrome (5 to 7 d) ($n = 12$)	COVID-19 (1 to 4 d) ($n = 26$)	COVID-19 (5 to 7 d) ($n = 26$)
Age, median (IQR) (years)	51.0 (21 to 79)	40.5 (21 to 73)	47.5 (20 to 89)	45 (27 to 67)
Female sex, n (%)	17 (54.8%)	13 (54.1%)	11 (33.3%)	16 (61.5%)
Comorbidities, n (%)	—	—	—	—
Cancer	0	1 (4.2%)	3 (9.1%)	0
Cardiovascular disease	2 (6.4%)	0	2 (6.1%)	2 (7.7%)
Chronic kidney disease	1 (3.2%)	0	2 (6.1%)	1 (3.8%)
Diabetes	1 (3.2%)	1 (4.2%)	5 (15.1%)	2 (7.7%)
High cholesterol	1 (3.2%)	0	0	0
Hypertension	8 (25.8%)	4 (16.7%)	9 (27.2%)	6 (23.1%)
Disease severity, n (%)	—	—	—	—
Mild	—	—	21 (63.6%)	19 (73.0%)
Moderate	—	—	4 (12.1%)	2 (7.7%)
Severe	—	—	5 (15.1%)	3 (11.5%)
Hospitalized	—	—	2 (6.1%)	2 (7.7%)
Death during hospitalization, n (%)	—	—	2 (6.1%)	0

Table 2. Clinical features of individuals from Brazil and Portugal with COVID-19.

	COVID-19 (Lisbon) (*n* = 11)	COVID-19 (Belo Horizonte) (*n* = 8)
Age, median (IQR) (years)	60 (45 to 85)	38 (35.7 to 56.8)
Female sex, n (%)	2 (18.1%)	3 (37.5%)
Comorbidities, n (%)	—	—
Cancer	0	0
Cardiovascular disease	1 (9.1%)	0
Choronic kidney disease	0	0
Diabetes	3 (27.3%)	0
High cholesterol	0	0
Hypertension	5 (45.4%)	0
Disease severity, n (%)	—	—
Mild	0	3 (37.5%)
Moderate	0	2 (25%)
Severe	11 (100%)	2 (25%)
Hospitalized	0	1 (12.5%)
Death during hospitalization, n (%)	3 (7.5%)	0

A different pattern was observed in individuals with FLS with a significant increase in the serum levels of CCL3, CCL4, IL-4, IL-15, TNF, IL-1B, IL-17A, FGF-basic, G-CSF, and GM-CSF after 4 d of infection (Fig. 1B to M). Interestingly, a reduction in IL-10 levels was observed over time (Fig. 1J). In patients with COVID-19, levels of cytokines such as TNF and IL1-RA were increased after 4 d of infection (Fig. 1F and I).

3.3 Patients with COVID-19, but not individuals with FLS, accumulated senescent/exhausted T cells shortly after infection

Since inflammatory response in critically ill patients with COVID-19 has been already associated with T-cell exhaustion,[24,28] we used flow cytometry immunophenotyping to evaluate alterations in immune responses during the first 7 d of infection to address whether this phenotype appeared in individuals with COVID-19 regardless of disease severity at an early stage. T-cell markers related to differentiation (CD28, CCR7, and CD45RO) and senescence/exhaustion (CD57, KLRG1, PD-1, TIGIT, and ICOS) allowed us to identify these phenotypes in distinct T-cell subsets including effector (CD28-CCR7-CD45RO−) and effector memory (EM) (CD28-CCR7-CD45RO+) T cells obtained by manual gating (Supplementary Fig. 1).

We observed a higher frequency of T CD8 and T CD4 cells expressing senescence and exhaustion markers (Fig. 2A to Q). However, in both subpopulations, an increase in cell senescence and exhaustion was detected after 4 d of the COVID-19 infection, a phenomenon that was not observed in patients with FLS. In patients with COVID-19 with 5 to 7 d of symptoms, there was an increase in the frequency of CD4+ KLRG1+ PD-1+ (Fig. 2C), CD4+ EFF PD-1+ (Fig. 2E); CD4+ EFF CD28-CD57+ KLRG1+ (Fig. 2F); CD4+ EM CD28-CD57+ ICOS+ (Fig. 2G); and CD4+ EM CD28-PD-1+ (Fig. 2H) cells. In the CD8 subpopulation, we observed differences in CD8 +CD57+ (Fig. 2I); CD8+ TIGIT+ ICOS+ (Fig. 2J); and CD8+ KLRG1+ PD-1+ (Fig. 2K) cells. Interestingly, no difference was observed in the frequency of effector cells (Fig. 2L), but an increase in the frequency of exhausted cells (CD8 EFF PD-1+; Fig. 2M) and senescent cells (CD8+ EFF CD28-CD57 + KLRG1+; Fig. 2N) was detected. There was also an increase in the frequency of EM CD8+ EM CD28-CD57 + ICOS + (Fig. 2O) and CD8+ EM CD28-PD-1 + exhausted/senescent cells (Fig. 2P). No difference was observed among patients with FLS ,suggesting again that the accumulation of these phenotypes after 4 d of symptoms was unique to SARS-CoV-2 infection.

A similar uniqueness of COVID-19 was observed when we examined frequencies of other phenotypes of effector and effector/memory CD4 and CD8 T cells expressing senescence/exhaustion markers (PD1, ICOS, CD57, and KLRG1) (Supplementary Fig. 2). Patients with COVID-19, but not FLS, with more than 4 d of symptoms had elevated frequencies of CD4 + EM CD28-CD57+ (Supplementary Fig. 2G), CD4 + EFF CD28-KLRG1+ (Supplementary Fig. 2P), CD8 + CD57 + PD-1+ (Supplementary Fig. 2B), CD8 + CD28 + ICOS+ (Supplementary Fig. 2E), CD8 + CD28 + PD1 + CD57 + KLRG1+ (Supplementary Fig. 2F), CD8+ EFF CD28-CD57+ (Supplementary Fig. 2J), CD8+ EFF CD28-CD57 + PD1+ (Supplementary Fig. 2M), CD8+ EM CD28-KLRG1+ (Supplementary Fig. 2N), CD8+ EM CD28-CD57 + PD1+ (Supplementary Fig. 2Q), and CD8+ EFF CD28-KLRG1+ (Supplementary Fig. 2R) when compared with patients with COVID-19 with 1 to 4 d of symptoms. No change was observed in the frequencies of regulatory T cells (Tregs) CD4 + CD25 + FOXP3+ (Supplementary Fig. 2S) or Tregs exhibiting an exhausted phenotype such as CD4 + CD25 + FOXP3 + PD-1+ (Supplementary Fig. 2T).

We also analyzed frequency of CD4 and CD8 T cells expressing different senescence and exhaustion makers (PD1, ICOS, CD28, KLRG1, TIGIT, and CD57) as well as the expression of the senescence and exhaustion markers in these cells using the mean fluorescence intensity (MFI) measurement (Supplementary Fig. 3). An increase in the expression of CD57 in CD4 + T cells was detected in the FLS group with 5 to 7 d of symptoms (Supplementary Fig. 3A), and the frequency of KLRG1+ CD8 T cells was also increased in individuals with more than 4 d of FLS symptoms. There was an augment in KLRG1+ CD8+ T cells in individuals with 5 to 7 d of COVID-19 symptoms with no changes in the expression of senescence or exhaustion markers (Supplementary Fig. 3B).

We also assessed the global frequencies of EFF (CCR7-CD45RO −), EM (CCR7-CD45R0+), and naïve (CCR7 + CD45RO−) CD4 and CD8 T cells in individuals with 1 to 4 and 5 to 7 d of symptoms. A reduction in the frequency of EM cells, an increase in Naïve cells in the CD4 T-cell population, and a reduction in EM cells in the CD8 T-cell population were observed, whereas EFF CD8 T cells were increased in individuals with 4 to 7 d of COVID-19 symptoms (Supplementary Fig. 4A and B).

In addition to changes in T-cell frequencies, changes in NK cells during COVID-19 infection were assessed. There was a decrease in the frequency of $CD56^{dim}$ NK cells, a population involved in IFN-γ production, CD56 + CD69− and cytotoxic NKG2D + NK cells in individuals with 5 to 7 d of COVID-19 symptoms (Supplementary Fig. 4C to H).

In view of these observations, we decided to evaluate the progression of exhaustion and senescence in the T-cell compartment during the first 7 d of COVID-19 symptomatic infection by conducting a longitudinal study in two different cohorts (Table 2).

3.4 Increased cellular senescence and exhaustion in patients with mild to moderate COVID-19 after 7 d of infection

To understand the immunological profile expressed by the cohorts of the longitudinal studies, we first compared the global production of inflammatory mediators between infected patients in time zero (T0) and 1 wk later at time seven (T7). Individuals from Belo Horizonte, Brazil, were younger and developed mild to moderate clinical symptoms of COVID-19 (Table 2). As expected, infected individuals showed a burst in inflammation throughout the analyzed period (Fig. 3a). When comparing T0 and T7, we observed a rise in the frequency of high producers

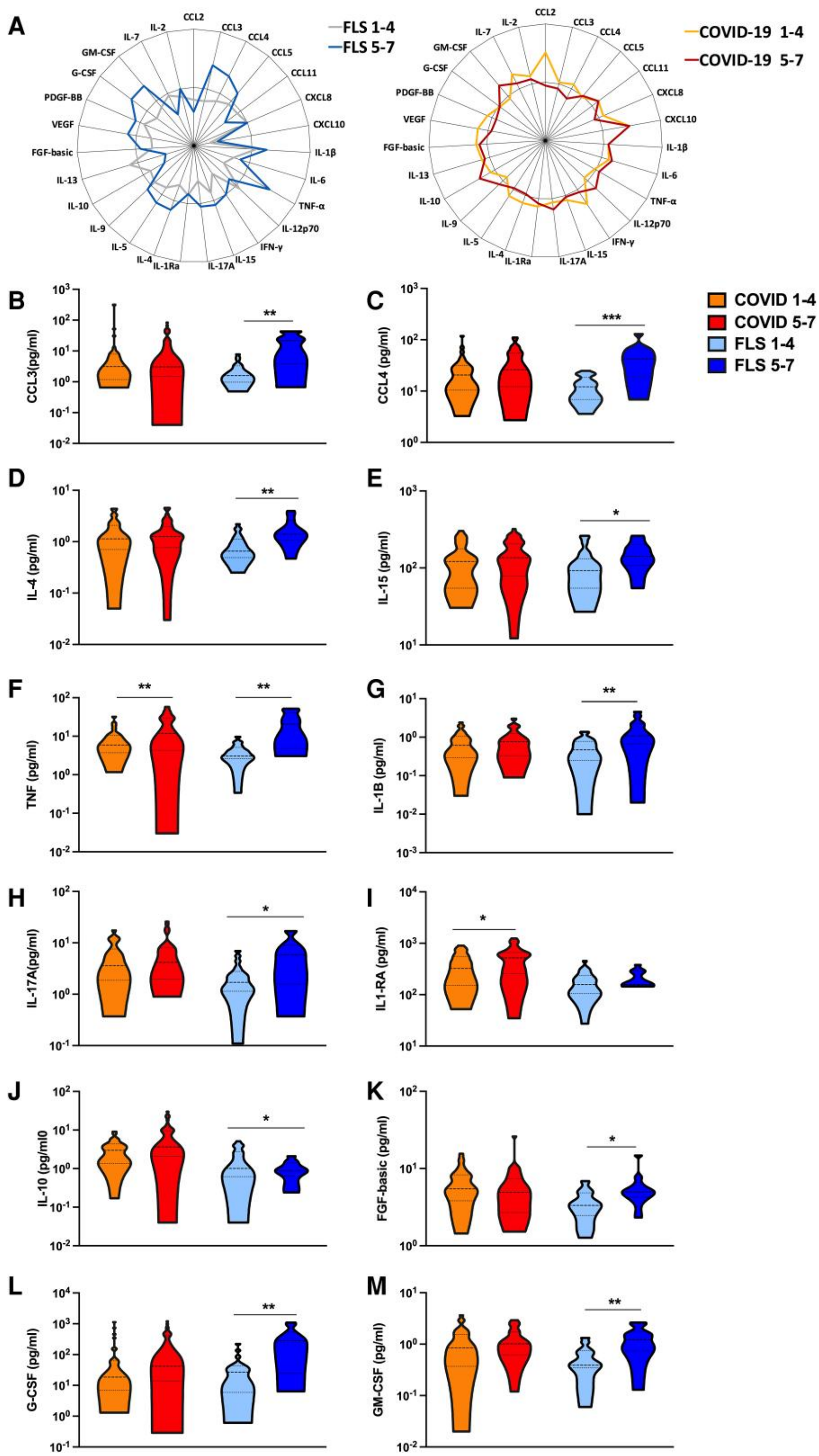

Fig. 1. Inflammatory profile of patients from Brazil with either FLS or COVID-19. A) Radar chart showing the baseline plasma cytokine/chemokine profile of individuals with either FLS or COVID-19 with 1 to 4 d of symptoms or with 5 to 7 d of symptoms. Radar chart represents the balance between inflammatory and antiinflammatory mediators from the innate and adaptive immunity compartments. B) Quantification of cytokines and other mediators in plasma obtained from patients with COVID-19 ($n = 52$) and FLS ($n = 43$). Data represent individual values, mean (center bar) ± SEM (upper and lower bars). Statistical analysis by two-sided Mann–Whitney nonparametric test; if not indicated, P value is not significant. Source data are provided as a Source Data file. Results considering $P < 0.05$.

of many cytokines, including CXCL-8, IL-1β, IL-2, IL-10, IL-5, IL-4, and IL-12p70, indicating intensification in the inflammatory response.

As for immunophenotyping analyses of individuals from Brazil, they presented an increase in the frequency of exhausted CD4+ T cells characterized as CD4 + PD-1+ (Fig. 3B)

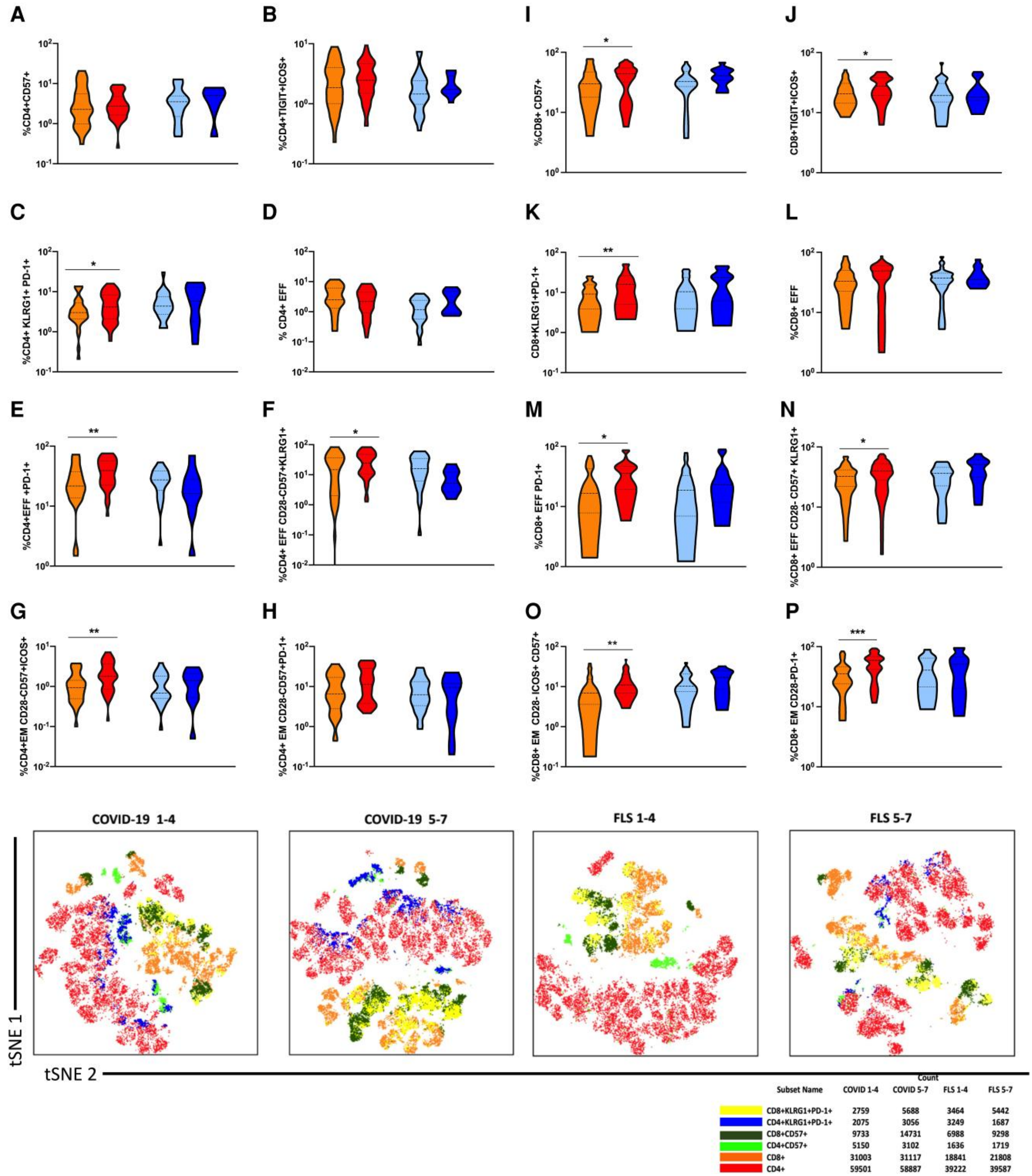

Fig. 2. Frequencies of exhausted/senescent T cells in patients from Brazil with either FLS or COVID-19. A) Representative tSNE plots of gated cell subsets and frequency analysis of either patients with FLS or COVID-19. B) to I) Frequencies of effector and effector/memory CD4 T cells expressing senescence/exhaustion markers PD1, ICOS, CD57, and KLRG1; J) and K) Frequencies of effector and effector/memory CD4 T cells expressing senescence/exhaustion markers (PD1, ICOS, CD57, and KLRG1). Statistical analysis was performed using the nonparametric Wilcoxon rank-sum test. $P < 0.05$.

and CD4+ CD28-TIGIT+ (Fig. 3D). In addition, CD4+ and CD8+ T cells expressing markers of senescence/exhaustion (CD57+ KLRG1+) in the EM T-cell compartment (CD45RO + CCR7− CD28−) were increased (Fig. 3C and F) after 7 d of symptoms. CD8+ T cells also presented an increase in effector cells with senescence/exhaustion markers CD8+ EFF CD28-CD57 + KLRG1+ (Fig. 3G).

We also observed an increase in both the frequency of CD28+ and PD1+ CD4+ T cells and in CD28+ CD8+ T cells as well as in the surface expression of CD28 by CD8+ T cells after 7 d of infection (Supplementary Fig. 5A and B).

Altogether, these results show a progressive increase in senescent/exhausted T-cell phenotypes in patients with mild to moderate COVID-19 during a short period of infection (7 d).

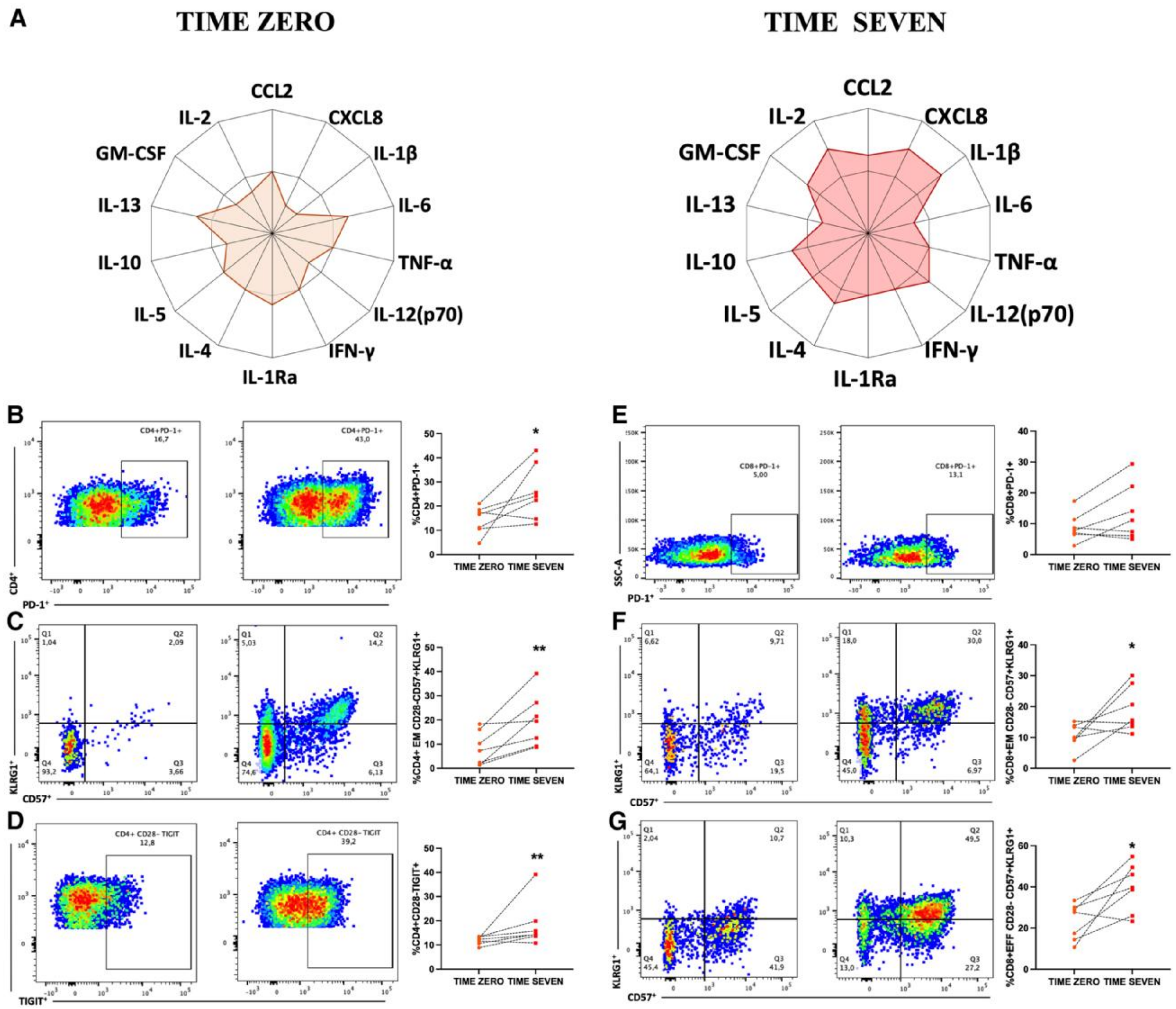

Fig. 3. Longitudinal study of inflammatory profiles of patients with COVID-19 from Brazil during a 7-d period of disease development. A) Radar chart representing the balance in frequencies of inflammatory and antiinflammatory cytokines/chemokines in the innate and adaptive immunity compartments from individuals between the time of recruitment (T0) at the hospital/ambulatory and 7 d later (T7) in the Brazilian cohort. B) to G) Representative flow cytometry plots of gated cell subsets and frequencies of CD4+ and CD8+ CD28-CD45RO+ CCR7− (EM) expressing senescence/exhaustion markers PD1, CD56, and KLRG1 obtained by manual gating in Brazilian patients at time points 0 and 7 of symptoms. Statistical analysis was performed using the nonparametric Wilcoxon rank-sum test. $P < 0.05$.

3.5 Severe COVID-19 increases cellular senescence and exhaustion in T cells

Severe COVID-19 has been reported to be associated with accelerated senescence.[22,23] Thus, we analyzed the cohort from Portugal that included individuals with COVID-19 classified as moderate and severe (Table 2). As presented in radar charts (Fig. 4A), infected individuals from Portugal exhibited an overall increase in the frequency of serum mediators in T0 when compared with the T7, confirming the ongoing response to the infection when they were recruited. Notably, there was an increase in the frequency of high producers of mediators involved in inflammatory response, such as IL-1β, GM-CSF, IL-5, and IL-12 over time (Fig. 4A). Seven days after hospitalization, the immune response was still more prominent than in noninfected individuals (data not shown), despite the reduction in the frequency of high producers of IL-4, CCL-2, IFN-γ, and TNF, possibly indicating a tendency toward resolution of the inflammatory response (Fig. 4A). Confirming this result, a significant reduction in the production of IFN-γ, IL 6, IL-10, and CCL-2, was also observed (Fig. 4B to E) after 7 d of infection.

Individuals from Lisbon/Portugal had higher average age and higher frequency of comorbidities, presenting greater COVID-19 severity when compared with individuals from Belo Horizonte/Brazil. These differences may explain a greater frequency of cells expressing exhaustion and senescence phenotypes amongst the former than the latter.

When evaluating the immunophenotyping profile, we analyzed the frequency of senescent cells using the markers CD57 and KLRG1. We identified an increased frequency of CD4+ KLRG1 + CD57+; CD8+ CD57+; and CD8+ KLRG1 + CD57+ (Fig. 4G, M, and N). Regarding the effector (EFF) and EM compartments, part of the transitional EM T-cell compartment is characterized by the expression of CD28 and the lack of CD45RO and CCR7. In addition, exhausted cells were analyzed by using the markers PD1 and ICOS. We identified an increase, after 7 d of severe COVID-19, in the frequency of senescent cells in the

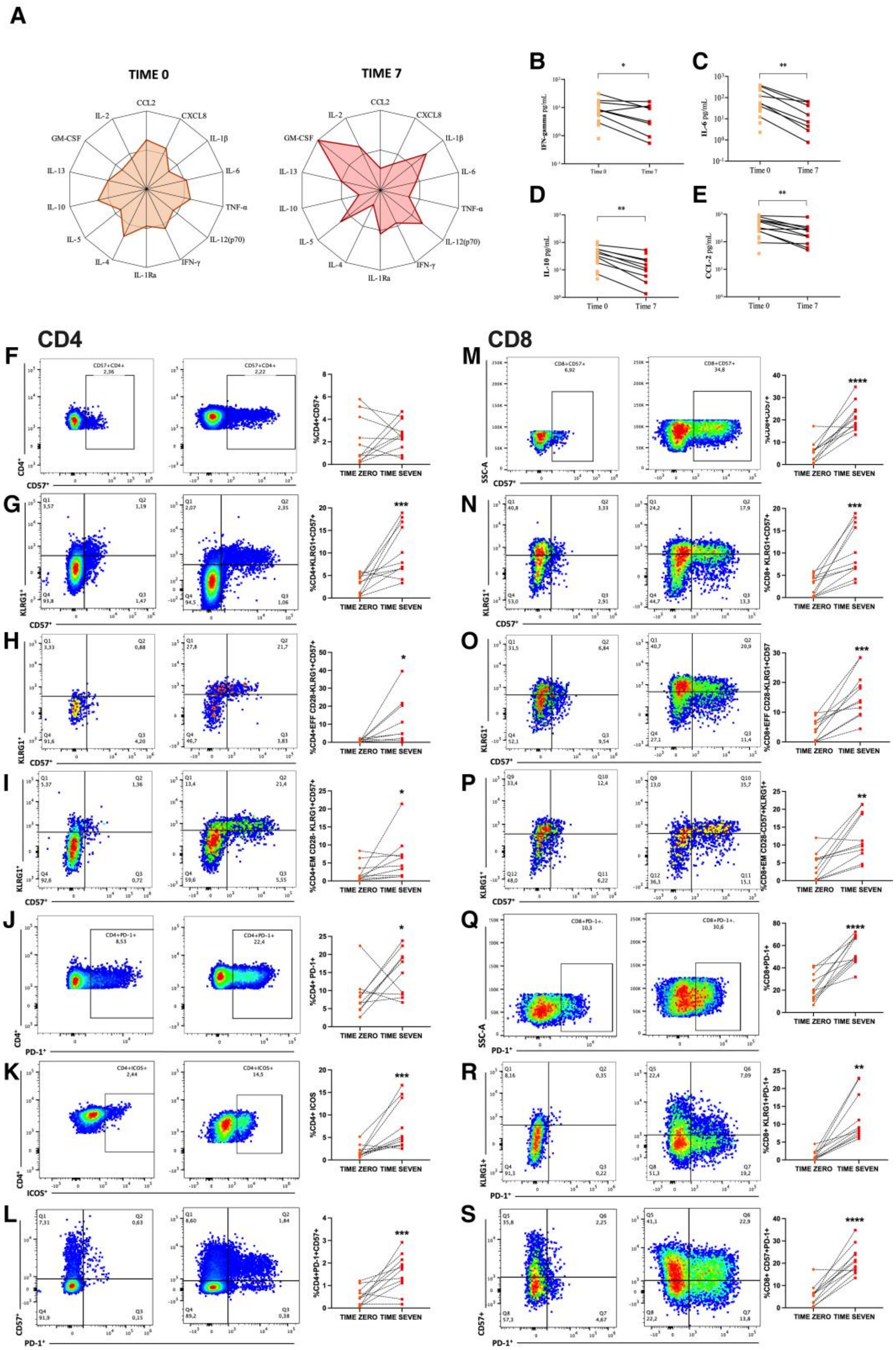

Fig. 4. Longitudinal study of inflammatory profiles of patients with COVID-19 from Portugal during a 7-d period of disease development. A) Radar chart of serum mediators of the control group (baseline) and patients with COVID-19 at hospitalization (T0) and 7 d later (T7) in the Portuguese cohort. B) to E) Comparison of serum levels of CCL-2, IFN-γ, IL-6, and IL-10 at hospital admission (T0) and 7 d later (T7). Wilcoxon's test was applied to compare samples of the same patient at T0 and T7. $P < 0.05$. F) to S) Representative flow cytometry plots of gated cell subsets and frequencies of naïve CD4 and CD8 T cells in individuals from Lisbon expressing senescence/exhaustion markers PD1, CD57, and KRLG1 obtained by manual gating. Statistical analysis was performed using the nonparametric Wilcoxon rank-sum test. $P < 0.05$.

effector T cell (CD4+ EFF CD28-KLRG1 + CD57+ and CD8+ EFF CD28-KLRG1 + CD57+) (Fig. 4H and O) and EM cell (CD4+ EM CD28-KLRG1 + CD57+ and CD8+ EM CD28-KLRG1 + CD57+) compartments (Fig. 4I and P) of both T CD4 and CD8 cells. In addition, we observed an increased frequency of exhausted T cells (CD4+ PD-1+; CD4+ ICOS+; CD8+ PD-1+; CD8+ ICOS+) (Fig. 4J, K, Q, and R) and a higher frequency of exhausted and senescent T cells (CD4+ PD-1 + CD57+ and CD8+ PD-1 + CD57+) (Fig. 4L and S).

Frequency of ICOS + CD4+ T cells increased, and expression of the exhaustion markers ICOS and PD1 increased in CD4+ T cells during the 7 d of COVID-19 infection (Supplementary Fig. 6A). In the CD8+ T-cell population, frequency of PD1+ cells also increased and the surface expression of PD-1, ICOS, and TIGIT was higher after 7 d of infection (Supplementary Fig. 6B).

4. Discussion

Inflammation induced by viral infections plays an important role in the differentiation and development of T cells, as they induce extensive proliferation, ultimately promoting cell exhaustion and senescence.[16,29,30] Although these changes are mainly related to chronic infections such as CMV, HIV, HBV, and HPV,[21,30] they have also been identified in patients with severe acute SARS-CoV2 infection.[24,31] Considering these findings and the highly heterogeneous immune responses observed in COVID-19,[32] our purpose was to evaluate whether mild/moderate forms of the disease or even other acute respiratory infections would have a similar impact in the T-, B-, and NK cell compartments. To address these questions, we compared the evolution of the inflammatory response and the exhaustion/senescence T-cell profile over a 7-d interval in a cross-sectional study in individuals with either COVID-19 or FLS. During the pandemic period, individuals with flu-like symptoms have sought out health units for testing. Therefore, we had the opportunity to compare patients with COVID-19 with the ones recruited at the same time and displaying similar symptoms but negative for SARS-CoV2 infection. These negative individuals became our control to respond to the question of whether COVID-19 had an effect on the immune system distinct from other respiratory viral infections.

Next, a longitudinal study with two other cohorts of patients with COVID-19 (one from Belo Horizonte/Brazil and one from Lisbon/Portugal) was performed to evaluate the dynamic change of T cells in individuals with mild or severe COVID-19.

COVID-19 is associated with the production of a large amount of inflammatory and antiinflammatory mediators, known as cytokine storm. In the early stage of the disease, common manifestations of acute inflammation, such as fever and headache, arise due to the exaggerated secretion of IL-1β, TNF, and IL-6. In addition to these, several other mediators, such as IFN-γ, IL-10, and IL-15, have been described in the inflammatory response to SARS-CoV2 (Lucas et al.[32]). Cytokine storm is not an exclusive phenomenon of COVID-19, and longitudinal studies conducted in patients with respiratory acute infections caused by other coronavirus[33]or influenza virus[34] showed elevated levels of cytokines such as IL-1β and IL-6 as well as chemokines such as IL-8, MCP-1 (CCL2), and IP-10 (CXCL-10) in the first week of infection.

Our data showed a distinct pattern of cytokines, chemokines, and other immune mediators in patients with COVID-19 during the first 7 d of infection when compared with patients who had flu-like symptoms and tested negative for SARS-CoV2, called here patients with FLS. We observed that the FLS group had an increase in the levels of serum mediators between 1 to 4 and 5 to 7 d of infection (Fig. 1B to M), whereas patients with COVID-19 already presented a very pronounced global production of serum mediators (over the 50% in the radar graph, Fig. 1A) since the first days of infection. In addition, the extent of global activation, measured by the area covered in the radar plot, is much higher in patients with COVID-19 than in patients with FLS, confirming the existence of an outburst of inflammatory mediators driven by SARS-CoV2 infection. This distinct pattern of inflammatory activation suggested that COVID-19 may have a unique impact on the differentiation of T cells at very early stages of the disease. To test this hypothesis, we searched for putative changes associated with exhaustion and senescence in different compartments of activated T cells during the 7-d period of infection.

T-cell differentiation into effector and memory cells occurs upon activation and may ultimately lead the cells toward exhaustion and senescence depending on the duration and strength of signaling. Exhaustion is characterized by the expression of high levels of inhibitory/differentiation receptors, including programmed cell death protein 1 (PD-1) and inducible co-stimulatory (ICOS). Expression of CD57 and killer cell lectin-like receptor G subfamily member 1 (KLRG1) and the loss of the co-stimulatory molecule CD28 are considered indicators of terminal differentiation and are associated with the senescence profile of T lymphocytes.[35,36] Although KLRG1 also emerges on the cell surface after differentiation, the signaling generated by this molecule regulates a pathway related to exhaustion, making it a marker of both cellular senescence and exhaustion.[35–39] Therefore, the absence of CD28 and the high expression of CD57 as well as KLRG1 provide what could be considered the best description of a senescent profile.[40] Exhausted and senescent T cells share common features, but they are two different functional states.[41] Exhausted T-cell dysfunction may be reversible, whereas senescence appears to be irreversible. The accumulation of senescent T cells increases over time, being higher in the elderly. Furthermore, in chronic viral infections or autoimmune disorders there is also an increase in both exhausted and senescent T cells.[42]

Notably, in the present study, we observed two major features of SARS-CoV2 infection regarding T-cell responses. First, COVID-19, but not FLS, was associated with a rapid increase in the frequency of senescent and exhausted T cells (Fig. 2, Supplementary Fig. 2). Second, this phenomenon was particularly prominent in CD8 T cells. Considering that T-cell activation can be influenced by numerous inflammatory cytokines produced by innate immune cells, the fact that SAR-CoV2 infection issued an outburst of inflammatory mediators at the first days of symptoms may favor the emergency of these cells during as early as 7 d. CD8 T cells are mostly involved in the immune response during COVID-19, being reactive to six different SARS-CoV2 epitopes.[43] In addition, a higher proportion of the CD8 T-cell responses to the viral spike protein were shown in individuals diagnosed with mild disease compared with those with severe disease, showing the importance of CD8 T cells in COVID-19 infection.[44] The high frequency of both senescent (CD57) and exhausted (PD-1) CD4 and CD8 cells in individuals with SARS-CoV2 infection analyzed in our study suggests that, differently from other respiratory viral infections, COVID-19 may rapidly compromise the immune responses to the virus and have a broad impact in the immune system. Exhausted T cells are at increased risk of apoptosis and have reduced responsiveness to antigens of the SARS-CoV-2, which can impair the ability to clear the infection.

There are distinct stimuli provided by viral infections that are drivers of both exhaustion and senescence. Usually, exhaustion is associated with persistent stimulation of the TCR by chronic viral infections or tumors. However, it has been suggested that the exhaustion program is distinct from the memory T-cell development, and it may be triggered at early stages of infection depending on the antigen signaling through the TCR and on the co-stimulation provided by other molecules.[45] The expression of the chromatin-associated protein TOX after TCR ligation seems to be essential for promoting exhaustion, especially in the setting of

CD8 T-cell stimulation.[46] On the other hand, cellular senescence is a stress-inducible cellular state distinct from exhaustion that can also be acutely triggered by SARS- CoV-2 infection. Virus-induced senescence has been detected in many viral infections including the ones caused by Retroviridae, Polyomaviridae, Paramyxoviridae, Parvoviridae, Rhabdoviridae, and Coronaviridae families, and it has been extensively shown in CMV and HIV infections and recently in SARS-CoV2 infection. SARS-CoV2 can induce cellular senescence via viral replication stress, reactive oxygen species production and cellular stress sensors such as Toll-like receptor 3 and cGAS–STING signaling.[47] These direct stimulations of both exhaustion and senescence in T cells may also be synergized by co-stimulatory events that amplify the differentiation program. One of the co-stimulatory stimuli that impact on several stages of T-cell differentiation is inflammatory mediators such as cytokines. It is known that inflammatory mediators present in the SASP developed by senescent cells such as IL-1, IL-8, and IL-6 can reinforce senescence in the neighboring cells.[48,49] This paracrine senescence works as a protective mechanism ensuring that critically damaged cells stop dividing. Activated T cells bear many receptors for cytokines. Therefore, it is plausible that several SASP-like mediators abundantly secreted during the cytokine storm provoked by COVID-19 play a critical role in boosting both exhaustion and senescence of T cells themselves.[50]

In addition to T cells, NK cells are an important subtype of innate lymphoid cells that play a role against viral infections. It has been previously demonstrated that human NK cells respond rapidly during the acute phase of human infections such as influenza A virus and dengue virus, as well as after vaccination with live attenuated yellow fever.[51] Leem et al. demonstrated that the CD56^{dim} NK population showed an expansion in the early phase of COVID-19 and are subsequently reduced throughout the infection regardless of disease severity, while the frequency of CD56^{bright} NK cells did not change. This typical change in NK cells during COVID-19 was accompanied by impaired NK cell cytotoxicity.[52] Another study in viral infection showed a reduction in the frequency of CD56^{dim} NK cells and decreased expression of NKG2D with increased production of IL-10 by CD56^{bright} NK cells in critically ill patients with hand, foot, and mouth disease, especially in EV71-(enterovirus A71)-infected patients.[53] Our findings also demonstrate a reduction in the frequencies of CD56^{dim} and cytotoxic NKG2D + NK cells throughout the infection. As already reported by others,[52] this result may reflect an initial expansion of activated NK cells to control the infection followed by a rapid contraction in their frequencies (in 7 d). Since our cohort is composed of individuals with different forms of COVID-19 (Table 1), the evolution in the frequencies of NK cells may not depend on disease severity either (Supplementary Fig. 4E to H).

Next, we designed a longitudinal study with two distinct cohorts of patients (one from Brazil and one from Portugal) with the following purposes: (i) to confirm the findings of the cross-sectional analysis during the early phase of COVID-19 and (ii) to clarify whether the rapid accumulation of exhausted/senescent T cells would differ between populations at different age groups, geographical locations, and with distinct forms of COVID-19 (mild/moderate vs. moderate/severe disease).

First, we observed that both cohorts had a hyperactivation of immune responses to SARS-CoV2 infection although they presented different inflammatory profiles (Figs. 3 and 4). Patients from Brazil with mild to moderate disease had a trend toward a global rise in serum cytokine production (Fig. 3A). For instance, the frequency of high producers of CXCL-8, a chemokine involved in the recruitment of neutrophils to the site of infection and commonly elevated in patients with COVID-19,[54–56] suggests an enhancement of the inflammatory response in the course of 7 d (Fig. 3A). In addition, we found higher frequencies of CCL-2 and GM-CSF levels in T7. These mediators are implicated in T lymphocyte response to SARS-CoV2, such as cell trafficking, cytokine production, and cell replication.[24,54]

A different scenario was observed with the severe patients from Portugal (Fig. 4) who exhibited high levels of biomarkers associated with a core response signature already described for moderate to severe conditions in COVID-19, such as IFN-γ, IL-6, and IL 10.[32,57] However, over the 7-d period, there was a trend toward the reduction of the inflammatory profile, as observed by the decrease in all mediator levels (Fig. 4a to E). These data are supported by the significant reduction of CCL2, IFN-γ, and IL-6 (Fig. 4b to E), which are biomarkers related to hyperactivation of T cells and recruitment of macrophages to the site of infection.[24,54] A decrease in the antiinflammatory cytokine IL-10 accompanies the general decline in inflammatory immune response indicating a trend to resolution of the inflammatory response.

It is possible that the two cohorts were not synchronized. In fact, we observed a more marked decrease in inflammatory cytokines in individuals from Lisbon over the 7-d period of infection. Although this heterogeneity may sound like a negative feature of our sample, the similar results yielded by the two distinct cohorts regarding the immunosenescence profile of the T-cell compartments helped to reveal a common phenomenon associated with the evolution of SARS-CoV2 infection.

Studies on individuals with severe and critically ill COVID-19 revealed an increase in T cells expressing markers of exhaustion/senescence such as PD-1, ICOS, and CD57, especially in the CD8 compartment, when compared with healthy individuals.[23–25,31] In the present work, our aim was to examine the emergence of these alterations in a longitudinal study and to determine whether they were restricted to severe cases of the disease in two distinct cohorts. Our data show that patients with mild and moderate cases of COVID-19 in Brazil had an increase in EM CD4+ CD28− T cells expressing both CD57 and KLRG1 (Fig. 3C to F), whereas patients from Lisbon with severe disease (Fig. 4I and P) show this increase in both CD4+ CD28− and CD8+ CD28− T-cell populations. In addition, individuals from Lisbon had an increase in the frequency of CD4+ and CD8+ T cells expressing PD-1. These changes in the T-cell compartment of Portuguese individuals with severe disease was accompanied by a similar augment in the expression of senescence (KLRG1, CD57) and exhaustion (PD1) markers in both CD4 and CD8 T cells (Supplementary Fig. 6). Therefore, our results identified the emergence of a senescence profile of T lymphocytes in individuals with COVID-19, in both mild and severe conditions, being more pronounced the more intense the inflammatory response to the disease was. Although there are no data in the literature relating inflammation and immunosenescence in COVID-19, the hyperactivation of the inflammatory response to SARS-CoV-2 infection has been associated with more severe disease,[32] which could possibly justify the increase in the frequency of terminally differentiated cells in this group of patients. It is also important to highlight that the cohort of critically ill patients from Portugal is mainly composed of elderly individuals. This may explain the differences in health conditions between Brazilian and Portuguese patients. As predicted by our group and others, immunosenescence and inflammaging may be aggravated by, but also may aggravate SARS-CoV-2 infection.[17,58] The primary setting and one of the hallmarks of immunosenescence is thymic involution, which is followed by the reduction in naïve cell output,[59] a phenomenon that should be affecting elderly patients even before the onset of infection. This could be the reason why no difference was observed in the naïve T-cell subset in these severely diseased elderly.

The differences seen in inflammatory biomarkers between patients of Brazil and Portugal may be also related to the time of the onset of symptoms, since the recruitment was carried out based on the moment patients sought medical assistance at the hospital, rather than on the initiation of symptoms. Notably, accumulation of exhausted/senescent cells seems to occur at distinct time points of disease development indicating a process that could start early in infection and develop further than the 7-d period examined, particularly in Brazil where the patients were recruited sooner in the course of infection.

There were limitations in this study, mainly the small sample size in the two cohorts of the longitudinal studies. There were also methodological differences in the study of the two cohorts that could have impacted the results. However, it confirmed and extended the findings of the cross-sectional study conducted on a larger population of Brazilians. Additionally, we did not investigate the SARS-CoV-2 variant infecting the individuals, which could have provided insights into the differences observed in immune responses, considering the distinct scenarios of the pandemic in Brazil and Portugal during which patients were recruited.

Our study provided evidence of accelerated immunosenescence in the T-cell compartment during COVID-19 showing that senescent and exhausted T cells significantly increased within a 7-d period. Interestingly, even though our samples were composed of two different populations, who presented distinct inflammatory profiles and were possibly affected by different variants of the virus, a similar profile of immunosenescence and exhaustion was identified during this short period of infection by SARS-CoV-2. Notably, accumulation of exhausted/senescent cells was not observed in patients with other flu-like infections and the changes in the immunological profile of individuals with COVID-19 were shown to be independent of disease severity even though the magnitude was greater in patients with severe disease. These findings suggest that accelerated immunosenescence of the T-cell compartment may represent a general and unique feature of COVID-19.

Acknowledgments

Samples in Lisbon were collected, stored, and provided by Biobanco-iMM (Lisbon Academic Medical Center, Lisbon, Portugal) in coordination with a taskforce created at iMM for the collection of COVID-19 research samples. Flow cytometry in Lisbon was performed at the Flow Cytometry Facility of iMM.

Author contributions

R.B.P. and L.T. helped designing the study, performed the experiments, and helped writing the manuscript. L.A.V. helped with patient recruitment and blood collection, helped performing the inflammatory profile experiments in Brazil, and helped writing the manuscript. G.C.C. helped with patient recruitment, clinical questionnaires, and flow cytometry experiments in Brazil and writing the manuscript. C. M. and A.C.M. helped with recruitment of patients and the experiments in Portugal. F.R. helped with phenotypic studies of patients in Portugal. H.C.G. supervised the recruitment of patients at Hospital Risoleta Neves in Brazil. R.C.B. performed the clinical examination of patients in Hospital Risoleta Tolentino Neves, Brazil. F.C. helped with the statistical analysis and data discussion for the manuscript writing. L.S.N. built the dataset containing clinical, phenotypical, inflammatory, biochemical, demographic, nutritional, serological data of all patients from Brazil. M.A.d.O. and V.D.M. did the follow-up of disease outcome of patients and helped with the phenotyping experiments in Brazil. G.S.-N. helped with statistical analysis and data discussion for the manuscript writing. U.T. supervised the recruitment of patients at UPA-CS in Brazil. A.T.C. supervised the experiments involving inflammatory and phenotypic profiling. L.G. and A.M.C.F. designed the study, coordinated all stages of the work, and wrote the manuscript.

Supplementary material

Supplementary material is available at *Journal of Leukocyte Biology* online.

Funding

This study was financially supported by grants from the Fundação para a Ciência e a Tecnologia (FCT, Portugal, Research4covid-369), Merck Sharp & Dohme (MISP 42521), and Conselho Nacional de Desenvolvimento Científico e Tecnológico (CNPq, Brazil, APQ 407363/2021-1). L.T., L.H.A.V., G.C.C., A.T.C., and A.M.C.F. are recipients of scholarships and fellowships from CNPq, Brazil. R.B.P. was funded by the FCT (SFRH/BD/144372/2019).

Conflict of interest statement. No conflict of interest.

References

1. Marshall JC, Murthy S, Diaz J, Adhikari N, Angus DC, Arabi YM, Baillie K, Bauer M, Berry S, Blackwood B, *et al.* A minimal common outcome measure set for COVID-19 clinical research. *Lancet Infect Dis*. 2020:20(8):e192–e197. https://doi.org/10.1016/S1473-3099(20)30483-7
2. Osuchowski MF, Winkler MS, Skirecki T, Cajander S, Shankar-Hari M, Lachmann G, Monneret G, Venet F, Bauer M, Brunkhorst FM, *et al.* The COVID-19 puzzle: deciphering pathophysiology and phenotypes of a new disease entity. *Lancet Respir Med*. 2021:9(6):622–642. https://doi.org/10.1016/S2213-2600(21)00218-6
3. Burki T. COVID-19 in Latin America. *Lancet Infect Dis*. 2020:20(5): 547–548. https://doi.org/10.1016/S1473-3099(20)30303-0
4. Thakur B, Dubey P, Benitez J, Torres JP, Reddy S, Shokar N, Aung K, Mukherjee D, Dwivedi AK. A systematic review and meta-analysis of geographic differences in comorbidities and associated severity and mortality among individuals with COVID-19. *Sci Rep*. 2021:11(1):8562. https://doi.org/10.1038/s41598-021-88130-w
5. Cifuentes-Faura J. COVID-19 mortality rate and its incidence in Latin America: dependence on demographic and economic variables. *Int J Environ Res Public Health Article*. 2021:18(13):6900. https://doi.org/10.3390/ijerph
6. Li B, Yang J, Zhao F, Zhi L, Wang X, Liu L, Bi Z, Zhao Y. Prevalence and impact of cardiovascular metabolic diseases on COVID-19 in China. *Clin Res Cardiol*. 2020:109(5):531–538. https://doi.org/10.1007/s00392-020-01626-9
7. Guo W, Li M, Dong Y, Zhou H, Zhang Z, Tian C, Qin R, Wang H, Shen Y, Du K, *et al.* Diabetes is a risk factor for the progression and prognosis of COVID-19. *Diabetes Metab Res Rev*. 2020:36(7): e3319. https://doi.org/10.1002/dmrr.3319
8. Zhou F, Yu T, Du R, Fan G, Liu Y, Liu Z, Xiang J, Wang Y, Song B, Gu X, *et al.* Clinical course and risk factors for mortality of adult inpatients with COVID-19 in Wuhan, China: a retrospective cohort study. *Lancet*. 2020:395(10229):1054–1062. https://doi.org/10.1016/S0140-6736(20)30566-3

9. Wu C, Chen X, Cai Y, Xia J, Zhou X, Xu S, Huang H, Zhang L, Zhou X, Du C, *et al.* Risk factors associated with acute respiratory distress syndrome and death in patients with coronavirus disease 2019 pneumonia in Wuhan, China. *JAMA Intern Med.* 2020:180(7):934–943. https://doi.org/10.1001/jamainternmed.2020.0994
10. Rahman A, Sathi NJ. Risk factors of the severity of COVID-19: a meta-analysis. *Int J Clin Pract.* 2021:75(7):e13916. https://doi.org/10.1111/ijcp.13916
11. Camell CD, Yousefzadeh MJ, Zhu Y, Langhi Prata LGP, Huggins MA, Pierson M, Zhang L, O'Kelly RD, Pirtskhalava T, Xun P, *et al.* Senolytics reduce coronavirus-related mortality in old mice. *Science.* 2021:373(6552):eabe4832. https://doi.org/10.1126/science.abe4832
12. Mueller AL, McNamara MS, Sinclair DA. Why does COVID-19 disproportionately affect older people? *Nat Aging.* 2020:12(10):9959–9981. https://doi.org/10.18632/aging.103344
13. Aiello A, Farzaneh F, Candore G, Caruso C, Davinelli S, Gambino CM, Ligotti ME, Zareian N, Accardi G. Immunosenescence and its hallmarks: how to oppose aging strategically? A review of potential options for therapeutic intervention. *Front Immunol.* 2019:10:2247. https://doi.org/10.3389/fimmu.2019.02247
14. Fulop T, Larbi A, Dupuis G, Le Page A, Frost EH, Cohen AA, Witkowski JM, Franceschi C. Immunosenescence and inflamm-aging as two sides of the same coin: friends or foes? *Front Immunol.* 2018:8:1960. https://doi.org/10.3389/fimmu.2017.01960
15. Hassouneh F, Campos C, López-Sejas N, Alonso C, Tarazona R, Solana R, Pera A. Effect of age and latent CMV infection on CD8+CD56+ T cells (NKT-like) frequency and functionality. *Mech Ageing Dev.* 2016:158:38–45. https://doi.org/10.1016/j.mad.2015.12.003
16. Heath JJ, Grant MD. The immune response against human cytomegalovirus links cellular to systemic senescence. *Cells.* 2020:9(3):766. https://doi.org/10.3390/cells9030766
17. Batista MA, Calvo-Fortes F, Silveira-Nunes G, Camatta GC, Speziali E, Turroni S, Teixeira-Carvalho A, Martins-Filho OA, Neretti N, Maioli TU, *et al.* Inflammaging in endemic areas for infectious diseases. *Front Immunol.* 2020:11:579972. https://doi.org/10.3389/fimmu.2020.579972
18. Elyahu Y, Monsonego A. Thymus involution sets the clock of the aging T cell landscape: implications for declined immunity and tissue repair. *Ageing Res Rev.* 2021:65:101231.
19. Goronzy JJ, Weyand CM. Mechanisms underlying T cell ageing. *Nat Rev Immunol.* 2019:19(9):573–583. https://doi.org/10.1038/s41577-019-0180-1
20. Solana R, Tarazona R, Aiello AE, Akbar AN, Appay V, Beswick M, Bosch JA, Campos C, Cantisán S, Cicin-Sain L, *et al.* CMV and immunosenescence: from basics to clinics. *Immun Ageing.* 2012:9(1):23. https://doi.org/10.1186/1742-4933-9-23
21. Fülöp T, Larbi A, Pawelec G. Human T cell aging and the impact of persistent viral infections. *Front Immunol.* 2013:4:271. https://doi.org/10.3389/fimmu.2013.00271
22. Diao B, Wang C, Tan Y, Chen X, Liu Y, Ning L, Chen L, Li M, Liu Y, Wang G, *et al.* Reduction and functional exhaustion of T cells in patients with coronavirus disease 2019 (COVID-19). *Front Immunol.* 2020:11:827. https://doi.org/10.3389/fimmu.2020.00827
23. Zheng HY, Zhang M, Yang CX, Zhang N, Wang XC, Yang XP, Dong XQ, Zheng YT. Elevated exhaustion levels and reduced functional diversity of T cells in peripheral blood may predict severe progression in COVID-19 patients. *Cell Mol Immunol.* 2020:17(5):541–543. https://doi.org/10.1038/s41423-020-0401-3
24. Arcanjo A, Guimarães Pinto K, Logullo J, Emílio Corrêa Leite P, Cristie Barreto Menezes C, Freire-de-Lima L, Diniz-Lima I, Decoté-Ricardo D, Nunes Rodrigues-da-Silva R, Geraldo Freire-de-Lima C, *et al.* Critically ill COVID-19 patients exhibit hyperactive cytokine responses associated with effector exhausted senescent T cells in acute infection. *J Infect Dis.* 2021:224(10):1672–1683. https://doi.org/10.1093/infdis/jiab425
25. Zheng M, Gao Y, Wang G, Song G, Liu S, Sun D, Xu Y, Tian Z. Functional exhaustion of antiviral lymphocytes in COVID-19 patients. *Cell Mol Immunol.* 2020:17(5):533–535. https://doi.org/10.1038/s41423-020-0402-2
26. Luiza-Silva M, Campi-Azevedo AC, Batista MA, Martins MA, Avelar RS, da Silveira Lemos D, Bastos Camacho LA, de Menezes Martins R, de Lourdes de Sousa Maia M, Guedes Farias RH, *et al.* Cytokine signatures of innate and adaptive immunity in 17DD yellow fever vaccinated children and its association with the level of neutralizing antibody. *J Infect Dis.* 2011:204(6):873–883. https://doi.org/10.1093/infdis/jir439
27. Cabaro S, D'Esposito V, Di Matola T, Sale S, Cennamo M, Terracciano D, Parisi V, Oriente F, Portella G, Beguinot F, *et al.* Cytokine signature and COVID-19 prediction models in the two waves of pandemics. *Sci Rep.* 2021:11(1):20793. https://doi.org/10.1038/s41598-021-00190-0
28. Lee S, Yu Y, Trimpert J, Benthani F, Mairhofer M, Richter-Pechanska P, Wyler E, Belenki D, Kaltenbrunner S, Pammer M, *et al.* Virus-induced senescence is a driver and therapeutic target in COVID-19. *Nature.* 2021:599(7884):283–289. https://doi.org/10.1038/s41586-021-03995-1
29. Moro-García MA, Mayo JC, Sainz RM, Alonso-Arias R. Influence of inflammation in the process of T lymphocyte differentiation: proliferative, metabolic, and oxidative changes. *Front Immunol.* 2018:9:492–499. https://doi.org/10.3389/fimmu.2018.00339
30. Wherry EJ. T cell exhaustion. *Nat Immunol.* 2011:12(6):492–499. https://doi.org/10.1038/ni.2035
31. De Biasi S, Meschiari M, Gibellini L, Bellinazzi C, Borella R, Fidanza L, Gozzi L, Iannone A, Lo Tartaro D, Mattioli M, *et al.* Marked T cell activation, senescence, exhaustion and skewing towards TH17 in patients with COVID-19 pneumonia. *Nat Commun.* 2020:11(1):3434. https://doi.org/10.1038/s41467-020-17292-4
32. Lucas C, Wong P, Klein J, Castro TBR, Silva J, Sundaram M, Ellingson MK, Mao T, Oh JE, Israelow B, *et al.* Longitudinal analyses reveal immunological misfiring in severe COVID-19. *Nature.* 2020:584(7821):463–469. https://doi.org/10.1038/s41586-020-2588-y
33. Wong CK, Lam CWK, Wu AKL, Ip WK, Lee NLS, Chan IHS, Lit LCW, Hui DSC, Chan MHM, Chung SSC, *et al.* Plasma inflammatory cytokines and chemokines in severe acute respiratory syndrome. *Clin Exp Immunol.* 2004:136(1):95–103. https://doi.org/10.1111/j.1365-2249.2004.02415.x
34. Gu Y, Zuo X, Zhang S, Ouyang Z, Jiang S, Wang F, Wang G. The mechanism behind influenza virus cytokine storm. *Viruses.* 2021:13(7):1362. https://doi.org/10.3390/v13071362
35. Akbar AN, Henson SM. Are senescence and exhaustion intertwined or unrelated processes that compromise immunity? *Nat Rev Immunol.* 2011:11(4):289–295. https://doi.org/10.1038/nri2959
36. Pangrazzi L, Weinberger B. T cells, aging and senescence. *Exp Gerontol.* 2020:134:110887. https://doi.org/10.1016/j.exger.2020.110887
37. Wherry EJ, Kurachi M. Molecular and cellular insights into T cell exhaustion. *Nat Rev Immunol.* 2015:15(8):486–499. https://doi.org/10.1038/nri3862
38. Cunha LL, Perazzio SF, Azzi J, Cravedi P, Riella LV. Remodeling of the immune response with aging: immunosenescence and its potential impact on COVID-19 immune response. *Front Immunol.* 2020:11:1748. https://doi.org/10.3389/fimmu.2020.01748

39. Pinti M, Appay V, Campisi J, Frasca D, Fülöp T, Sauce D, Larbi A, Weinberger B, Cossarizza A. Aging of the immune system: focus on inflammation and vaccination. *Eur J Immunol*. 2016:46(10): 2286–2301. https://doi.org/10.1002/eji.201546178
40. Rodriguez IJ, Lalinde Ruiz N, Llano León M, Martínez Enríquez L, Montilla Velásquez M del P, Ortiz Aguirre JP, Rodríguez Bohórquez OM, Velandia Vargas EA, Hernández ED, Parra López CA. Immunosenescence study of T cells: a systematic review. *Front Immunol*. 2021:11:604591. https://doi.org/10.3389/fimmu.2020.604591
41. Zhao Y, Shao Q, Peng G. Exhaustion and senescence: two crucial dysfunctional states of T cells in the tumor microenvironment. *Cell Mol Immunol*. 2020:17(1):27–35. https://doi.org/10.1038/s41423-019-0344-8
42. Akbar AN, Henson SM, Lanna A. Senescence of T lymphocytes: implications for enhancing human immunity. *Trends Immunol*. 2016:37(12):866–876. https://doi.org/10.1016/j.it.2016.09.002
43. Kared H, Redd AD, Bloch EM, Bonny TS, Sumatoh H, Kairi F, Carbajo D, Abel B, Newell EW, Bettinotti MP, *et al*. SARS-CoV-2-specific CD8+ T cell responses in convalescent COVID-19 individuals. *J Clin Invest*. 2021:131(5):e145476. https://doi.org/10.1172/JCI145476
44. Peng Y, Mentzer AJ, Liu G, Yao X, Yin Z, Dong D, Dejnirattisai W, Rostron T, Supasa P, Liu C, *et al*. Broad and strong memory CD4+ and CD8+ T cells induced by SARS-CoV-2 in UK convalescent individuals following COVID-19. *Nat Immunol*. 2020:21(11): 1336–1345. https://doi.org/10.1038/s41590-020-0782-6
45. Blank CU, Haining WN, Held W, Hogan PG, Kallies A, Lugli E, Lynn RC, Philip M, Rao A, Restifo NP, *et al*. Defining 'T cell exhaustion'. *Nat Rev Immunol*. 2019:19(11):665–674. https://doi.org/10.1038/s41577-019-0221-9
46. Khan O, Giles JR, McDonald S, Manne S, Ngiow SF, Patel KP, Werner MT, Huang AC, Alexander KA, Wu JE, *et al*. TOX transcriptionally and epigenetically programs CD8+ T cell exhaustion. *Nature*. 2019:571(7764):211–218. https://doi.org/10.1038/s41586-019-1325-x
47. Schmitt CA, Tchkonia T, Niedernhofer LJ, Robbins PD, Kirkland JL, Lee S. COVID-19 and cellular senescence. *Nat Rev Immunol*. 2023:23(4):251–263. https://doi.org/10.1038/s41577-022-00785-2
48. Acosta JC, O'Loghlen A, Banito A, Guijarro MV, Augert A, Raguz S, Fumagalli M, Da Costa M, Brown C, Popov N, *et al*. Chemokine signaling via the CXCR2 receptor reinforces senescence. *Cell*. 2008:133(6):1006–1018. https://doi.org/10.1016/j.cell.2008.03.038
49. Kuilman T, Michaloglou C, Vredeveld LCW, Douma S, van Doorn R, Desmet CJ, Aarden LA, Mooi WJ, Peeper DS. Oncogene-induced senescence relayed by an interleukin-dependent inflammatory network. *Cell*. 2008:133(6):1019–1031. https://doi.org/10.1016/j.cell.2008.03.039
50. Tsuji S, Minami S, Hashimoto R, Konishi Y, Suzuki T, Kondo T, Sasai M, Torii S, Ono C, Shichinohe S, *et al*. SARS-CoV-2 infection triggers paracrine senescence and leads to a sustained senescence-associated inflammatory response. *Nat Aging*. 2022:2(2):115–124. https://doi.org/10.1038/s43587-022-00170-7
51. Abel AM, Yang C, Thakar MS, Malarkannan S. Natural killer cells: development, maturation, and clinical utilization. *Front Immunol*. 2018:9:1869. https://doi.org/10.3389/fimmu.2018.01869
52. Leem G, Cheon S, Lee H, Choi SJ, Jeong S, Kim ES, Jeong HW, Jeong H, Park SH, Kim YS, *et al*. Abnormality in the NK-cell population is prolonged in severe COVID-19 patients. *J Allergy Clin Immunol*. 2021:148(4):996–1006.e18. https://doi.org/10.1016/j.jaci.2021.07.022
53. Pan Z, Zhao R, Shen Y, Liu K, Xue W, Liang C, Peng M, Hu P, Chen M, Xu H. Low-frequency, exhausted immune status of CD56dim NK cells and disordered inflammatory cytokine secretion of CD56bright NK cells associated with progression of severe HFMD, especially in EV71-infected patients. *Int Immunopharmacol*. 2021:101:108369. https://doi.org/10.1016/j.intimp.2021.108369
54. Coperchini F, Chiovato L, Ricci G, Croce L, Magri F, Rotondi M. The cytokine storm in COVID-19: further advances in our understanding the role of specific chemokines involved. *Cytokine Growth Factor Rev*. 2021:58:82–91. https://doi.org/10.1016/j.cytogfr.2020.12.005
55. Sallenave JM, Guillot L. Innate immune signaling and proteolytic pathways in the resolution or exacerbation of SARS-CoV-2 in COVID-19: key therapeutic targets? *Front Immunol*. 2020:11:1229. https://doi.org/10.3389/fimmu.2020.01229
56. Hazeldine J, Lord JM. Neutrophils and COVID-19: active participants and rational therapeutic targets. *Front Immunol*. 2021:12: 680134. https://doi.org/10.3389/fimmu.2021.680134
57. Liu J, Li S, Liu J, Liang B, Wang X, Wang H, Li W, Tong Q, Yi J, Zhao L, *et al*. Longitudinal characteristics of lymphocyte responses and cytokine profiles in the peripheral blood of SARS-CoV-2 infected patients. *EBioMedicine*. 2020:55:102763. https://doi.org/10.1016/j.ebiom.2020.102763
58. Witkowski JM, Fulop T, Bryl E. Immunosenescence and COVID-19. *Mech Ageing Dev*. 2022:204:111672. https://doi.org/10.1016/j.mad.2022.111672
59. Thomas R, Wang W, Su DM. Contributions of age-related thymic involution to immunosenescence and inflammaging. *Immun Ageing*. 2020:17(1):2. https://doi.org/10.1186/s12979-020-0173-8